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Sequential Clustering and Classification Approach to Analyze Sales Performance of Retail Stores Based on Point-of-Sale Data

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Point-of-Sale (POS) data analysis is usually used to explore sales performance in business commence. This manuscript aims to combine unsupervised clustering and supervised classification methods in an integrated data analysis framework to analyze the real-world POS data. Clustering method, which is performed on sales dataset, is used to cluster the stores into several groups. The clustering results, data labels, are then combined with other information in store features dataset as the inputs of the classification model which classifies the clustering labels by using store features dataset. Non-dominated sorting generic algorithm-II (NSGA-II) is applied in the framework to employ the multi-objective of clustering and classification. The experimental case study shows clustering results can reveal the hidden structure of sales performance of retail stores while classification can reveal the major factors that effect to the sales performance under different group of retail stores. The correlations between sales clusters and the store information can be obtained sequentially under a series of data analysis with the proposed framework.

Keywords: Data Clustering; data classification; multi-objective optimization; non-dominated sorting genetic algorithm; point of sale data.

1. Introduction

Point-of-sale (POS) data are commonly used for sales performance analysis, for example, analyzing sales effects, [1] improving customer service, [2] and sale prediction. [3,4] The transactional data contain statistical information that identifies a particular customer who bought a set of products at certain prices, at a store or branch, and

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time.[5,6] Fisher and Raman also mentioned that the proper analysis of sales data can be extremely useful to a retailer in improving decision-making on optimizing store assortments and dynamic pricing of items. [7] Understanding the POS data not only reveals the customers' purchase behaviors but also helps in enhancing the supply chain management with more e®ective inventory control. For instance, POS data analysis can discover how well an item was sold. Then, using this information, a retailer can adjust the ordering level correspondingly as well as make pricing decisions that adapt the trend and seasonality of the item at di®erent periods. [8]

Duchessi *et al.* grouped the existing techniques used for the retail POS data analysis into three approaches [9]: (1) query-based method, (2) statistical techniques, and (3) data-mining or machine learning method. Recently, Choi *et al.* also summarized that statistics, machine learning, data mining, and optimization are the main big data analytics techniques that have been utilized in operation management areas.[10] Essentially, the query-based methods by database query language (SQL) are evaluated as easy-to-use techniques on the datasets with the prede¯ned columns/features. However, these techniques were unable to detect hidden information and knowledge. The statistical techniques such as multiple analysis of variance (MANOVA), cluster analysis, regression analysis, and factor analysis can perform multivariate patterns' analysis of association discovery. Although statistical techniques provide scienti¯c interference, it is hard to interpret the results under multidimensional datasets in a big data environment as well as to construct a statistical approach when the assumption of data distribution might not be held. The algorithms of data mining or machine learning techniques such as tree induction, neural network induction, and association rule induction can provide promising analytics on a relatively higher level of data complexity in large-scale data than the other approaches. Conversely, the \algorithm-based" approach requires professional model building and the coding of data into classes to handle the big data.

In general, a framework that can conduct data mining functions simply and straightforwardly to analyze POS data is necessary for daily or weekly retailing analysis. For example, a manager who is in charge of supervising multiple stores or branches would like to investigate the sales performance of each store and also to search out which factors in°uence the sales performance. To conduct the analysis, usually, the performance of each store can be represented by the revenue received within a speci¯ed period, such as a season, month, or week. To di®erentiate the store performance, a grouping technique or clustering method is required to cluster the stores by their multivariate time series revenue data. Based on the clustering result, a manager would like to ¯nd out if there is a special sales pattern existing in the clustering results. Besides, the manager might want to know further why those stores have similar sales performances. Is this similarity based on store geometry, location, customer demographics, or local competition? In other words, revealing the factors related to the store similarity in terms of sale performance among stores can help in managing the assortment of the item in similar stores (\clusters") rather than trying to deal with listing items each store or each item individually. Factors regarding the

stores such as location, human resources, population around the store, available transportation nearby the store, and so on are critical in in°uencing the sales performance. Thus, utilizing classi¯cation analysis to ¯nd the most correlated factors which a®ect the sales signi¯cantly is important.

Based on the analytics scenarios addressed beforehand, a systematic data analysis framework is needed to handle the daily or weekly analysis tasks as a routine. In this research, a data mining framework combining data clustering and classi¯cation tasks using fast and elitist non-dominated sorting genetic algorithm (NSGAII-SCC) [11] is employed to analyze POS data. Clustering stores or sale items by the sale records help in revealing the hidden information behind the selling pattern. Classi¯cation analysis is applied to identify the factors which are likely correlated with the found sales pattern in each cluster. By combining clustering and classi¯cation on data exploration, an interesting research question was raised. When utilizing the clustering method on the performance measures and then further using the classi¯cation method to reveal the correlated factors, how to maintain the clustering and classi¯cation e®ectiveness (or qualities) to show the con¯dence of the POS data analysis? The complexity of this combined analysis is depended upon the number of ways to cluster retail stores and the multiple possibilities of the associated factors related to the found store clusters (in terms of sale performance), for each clustering case. As studied in the literature, the clustering problem is an NP-hard optimization problem,[12] and searching the in°uential factor so-called the feature selection for classi¯cation is also a complicated technique, especially for a large dataset. [13] When searching space of ¯nding the correlation among the in°uential factors and retail performance by POS data is large, how to conduct the data analysis under the need of investigating retailing performance is an interesting research question.

To attack this research question, the NSGA-II-SCC method simultaneously optimizes the solution quality of clustering and classi¯cation. The objective functions are formed as (1) to minimize the clustering compactness when performing store clustering based on the revenue performance and (2) to maximize the classi¯cation accuracy when investigating factors that likely contribute to classifying the revenue performance of each store cluster. For this combined data analysis, the number of clusters (k), which is a user-de¯ned parameter, seems crucial and needs further investigation. Too many clusters may lead to classi¯cation di±culty in recognizing the correlated factors and loss of generality. On the other hand, too few clusters may make the partitions too vague that the classi¯cation result may be too general and lose the investigation capability. Thus, this study uses Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) [14] method to analyze the Pareto-optimal solutions to determine the best number of clusters with its corresponding solution for the framework. Besides, stepwise regression technique [15] is employed to identify the signi¯cant features for classi¯cation which are highly correlated with the store patterns.

This paper is structured as follows. Section 2 reviews POS data analysis and clustering and classi¯cation techniques. Section 3 describes the methodology of the proposed framework. Section 4 shows the experimental result of applying the

proposed framework on the POS data of a bakery retail chain as a case study. Section 5 presents the conclusion and future research direction.

2. Literature Review

2.1. *POS data analysis*

The POS system contains information on sold products such as the item number, price, selling time, location, and customer information. Sales data are convenient to be collected and managed using a POS system and have been applied widely and e®ectively on di®erent commercial activities. The POS data will be stored and analyzed so that the inventory and supply information can be provided rapidly to decrease labor costs and improve productivity. For example, Croson and Donohue studied the impact of sharing POS data on reducing the bullwhip e®ect in supply chain management. [16] Sharing POS information increased supply chain performance by decreasing the magnitude of order oscillations for manufacturers, distributors, wholesalers, and retailers. Karen Stein analyzed POS data for food service and built a system to expedite customer service and support the managers to control their business such as who are dining, what is selling as well as the inventory. [17] Li *et al.* analyzed ¯nancial data to de¯ne the groups of customers' behaviors and the potential risks. In their study, a new cluster validation index and a novel penalty-function-based solver were proposed to automatically detect customer groups as well as re¯ne the clusters' centroids and hyperellipsoidal scopes. [18]

In the literature, using POS data to predict product sales is one of the popular research works in POS data analysis. POS data can be processed in di®erent ways for sales forecasting, which will bring great e®ects on marketing and sales strategies. [3] POS data are formed in chronological order, as a result, time series models are commonly used. Sundararaman *et al.* adopted the POS data and considered the databases of trends, seasonality, and seasonal index using centered moving averages and normalization. [4] Holt-Winters model used the original trend and seasonal factor to calculate the average value of POS data. Employing deleting and ¯ltering the seasonal factor, fewer forecasting errors and better forecasting results can be presented. Williams *et al.* also aggregated order forecasting and past order information, executing processes and other uncertain factors from POS data. [19] Their model ¯ltered the in°uence and errors of the bullwhip e®ect and proposed a long-term balance of inventory strategy.

Some researchers studied the e®ect of customers' behaviors based on POS data. Duchessi *et al.* used POS data to group retail grocery chain customers based on their similar purchasing behavior by geodemography. [9] Kashima *et al.* developed a recommendation method that integrated a POS system and an automatic order function for a restaurant. [20] The developed system recommended a menu to customers based on the customers' pro¯le analyzed based on POS data. Ogawa *et al.* used POS data to analyze customers' purchasing behaviors on vegetables and fruits. [2] The sales data revealed that point-of-purchase health information could be e®ective to foster

customers' healthy dietary habits. Aloysius *et al.* also investigated the service process perceptions to shipping outcomes in the retail environment by using \mobile" POS data carried by the smartphone. [21] They found that the technology enablers on mobile POS systems in scanning and payment scenarios give retailers competitive advantage in the era of big data. All of the mentioned research utilized certain data analysis methods or analysis framework to conduct the study under POS data.

Although POS data have been used in a variety of research domains, a few research, based on our knowledge, focuses on developing the particular data mining technique for analyzing POS data to deal with the analysis of retailing performance and the corresponding factors. Therefore, this study aims to develop a new data analysis framework that combines data clustering and data classi¯cation to analyze POS data. The proposed technique is expected to apply to general POS datasets. In this work, a case study of a bakery retail store chain was demonstrated as an example of the utilization of the developed framework.

2.2. *Review of data clustering and data classi¯cation*

To cluster retail stores by the revenue performance and further investigate the in°uential factors, the clustering and feature selection for classi¯cation methods are commonly applied on analyzing POS data. Data clustering and data classi¯cation are the two most popular data mining techniques. The clustering method is mostly known as an unsupervised learning method because clustering is performed without available information that is related to the membership of data items to predetermined labels. [22] Soni and Ganatra explained that clustering is exploratory in nature and the objective is to ¯nd structure in a dataset. [23] The clustering process categorizes similar objects of data into a group called a cluster. The similarity measure introduces the compactness of clusters as a performance measure for clustering. Generally, clustering algorithms are divided into two types: hierarchical clustering and partitioning clustering. The hierarchical clustering approach analyzes the dataset into a hierarchy and typically presents the results in dendrograms or trees. The partitioning clustering such as k-means method divides large data into a smaller number of groups based on similarity. To evaluate the clustering result, some cluster validation indices are used such as accuracy, rand index, entropy, purity, and so on. Besides, Kou *et al.* employed multi-criteria decision making (MCDM) method to evaluate clustering performance. [24]

Contrary to data clustering, data classi¯cation methods give classes or labels to new data records based on the existing data. [22] There are two steps in data classi¯-cation. The ¯rst step is to create a model with the given data, also known as the training data. The second step is to use the created model to predict the class label for the incoming unknown data based on the selected features. The complexity of these steps depends on the number of features to use as classi¯ers. Some popular classi¯cation methods before deep learning methods such as decision tree (DT), arti¯cial neural network (ANN), k-nearest neighbor (KNN), and support vector machines (SVM) have been used widely, [25] and recently emerging deep learning

method is popular. [26] Commonly, clustering and classi¯cation are implemented separately with di®erent models. Some researchers combined clustering and classi¯cation to improve the performance of the target task. For instance, some studies used clustering to improve classi¯cation results such as (1) combining *k*-means with decision tree learning, [27] (2) clustering ensemble with SVM, [28] (3) *k*-means with SVM and regression, [29] (4) multiple clustering approaches such as *k*-means, fuzzy *c*-means, and hierarchical clustering with SVM and ANN, [30] and (5) ensemble classi¯cation-based supervised clustering (ECSC). [31] Besides, some research works considered applying classi¯cation methods to cluster the classi¯ed results. Those studies include a combination of *k*-means and SVM in supervised *k*-means clustering method (SSVM), [32] and integration of SVM classi¯er with multi-objective genetic algorithm (MOGA) clustering. [33] Moreover, multiple studies try to perform clustering and classi¯cation simultaneously or sequentially in one model to utilize both data mining methods. Cai *et al*. proposed a complicated multi-objective simultaneous learning framework for clustering and classi¯cation (MSCC) in which both clustering and classi¯cation models can rely on the clusters' centers. [34] Bian *et al*. proposed a joint clustering and classi¯cation technique for remote-sensing images with bipartition-based *k*-means and SVM classi¯er combined. [35] To simply present the results or clustering and classi¯cation, Yang and Quyen proposed a sequential clustering and classi¯cation data analysis framework to consider clustering one dataset and classifying the clustering result by another dataset to reveal the correlation between the two datasets. [11] To handle the multi-objective optimization, in literature, multi-objective genetic algorithm (MOGA) is suggested. There are multiple versions of MOGA such as vector evaluated genetic algorithm (VEGA), [36] non-dominated sorting genetic algorithm (NSGA), [37] fast and elitist non-dominated sorting genetic algorithm (NSGAII), [38] and so on. Due to superiority of multi-objective optimization, in Yang and Quyen's work, a fast and elitist non-dominated sorting genetic algorithm (NSGAII) was used to combine clustering with classi¯cation in a learning framework to ensure objective functions of clustering and classi¯cation complement each other.

Although NSGAII-SCC has the straightforward framework of the sequential approach and promising performance, how to conclude the relationships between the results of clustering and classi¯cation based on the multiple solutions generated from the Pareto front is needed for POS data analysis. In this research, TOPSIS and the stepwise regression analysis were applied to extend the original NSGAII-SCC framework to resolve the solution selection issue. The detailed information regarding the extension of TOPSIS and the stepwise regression analysis can be found in the following section.

3. Methodology

In this study, sequential clustering and classi¯cation are utilized to analyze the performance of a retail chain based on the store sales patterns. The framework of sequential clustering and classi¯cation using a fast and elitism non-dominated

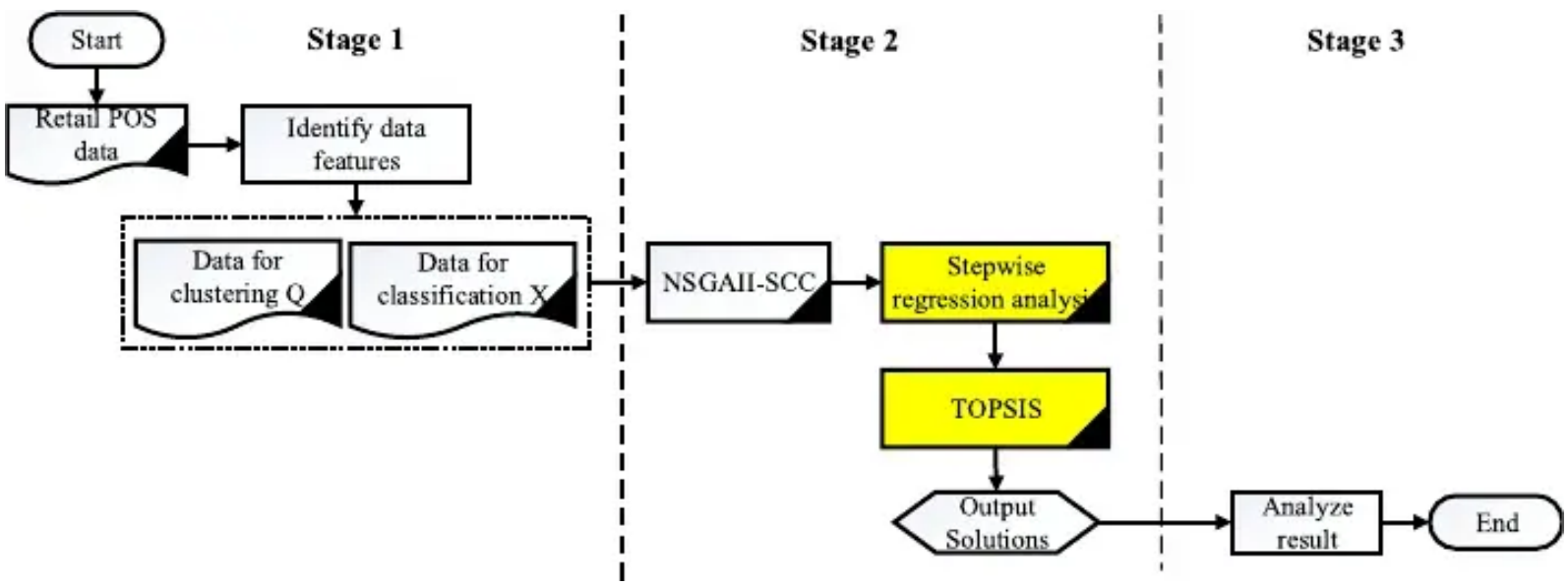


Fig. 1. Framework of NSGAII-SCC on retail POS data with the stepwise regression analysis for feature selection and TOPSIS for solution selection.

sorting genetic algorithm (NSGAII-SCC) [11] for POS data analysis is illustrated in Fig. 1. Essentially, there are three stages in this framework. In the ¯rst stage, the retail POS data are processed to identify and separate data features into two di®erent datasets: sales dataset (Q dataset) for clustering and store features dataset (X dataset) for classi¯cation. The second stage performs clustering and classi¯cation sequentially on the two datasets which are identi¯ed in the ¯rst stage. Herein, clustering on the Q dataset is ¯rst performed to exploit the data patterns. Then, the class labels obtained from the clustering result are used for training the classi¯er on the X dataset. Besides, NSGAII is employed in the second stage to iteratively search for the optimal solution. In addition, compared with the original NSGAII-SCC, [11] an extension of data analysis was proposed in this stage to evaluate the feature mechanism by stepwise regression method. Essentially, the chromosomes of each generation of sequential clustering and classi¯cation are recorded. Then, a stepwise regression model is applied to eliminate the redundant and non-signi¯cant features for classi¯cation based on the multiple solutions. Finally, the third stage analyzes the output results which reveal the data patterns and identify the correlated features to the sale patterns.

3.1. *Stage 1: Data features identi¯cation*

POS data contain transactional records of sales which usually consist of two parts of information. One is sale records containing date, time, items, transaction number, prices, and so on, as the Q dataset mentioned in Fig. 1. Another part of data is information regarding the store features such as store identi¯cation number, store geometry size, and region information regarding the stores as X dataset mentioned in Fig. 1. These data are saved in a POS system and are retrievable to analyze sales patterns, pro¯t, number of customers, and others.

The sale transaction data are usually recorded in time series data format in POS. The data can be aggregated daily, weekly, monthly, or annually based on the research purpose. In this study, the sales data are aggregated monthly by store identi¯cation number. Please note that the aggregation can be conducted in di®erent

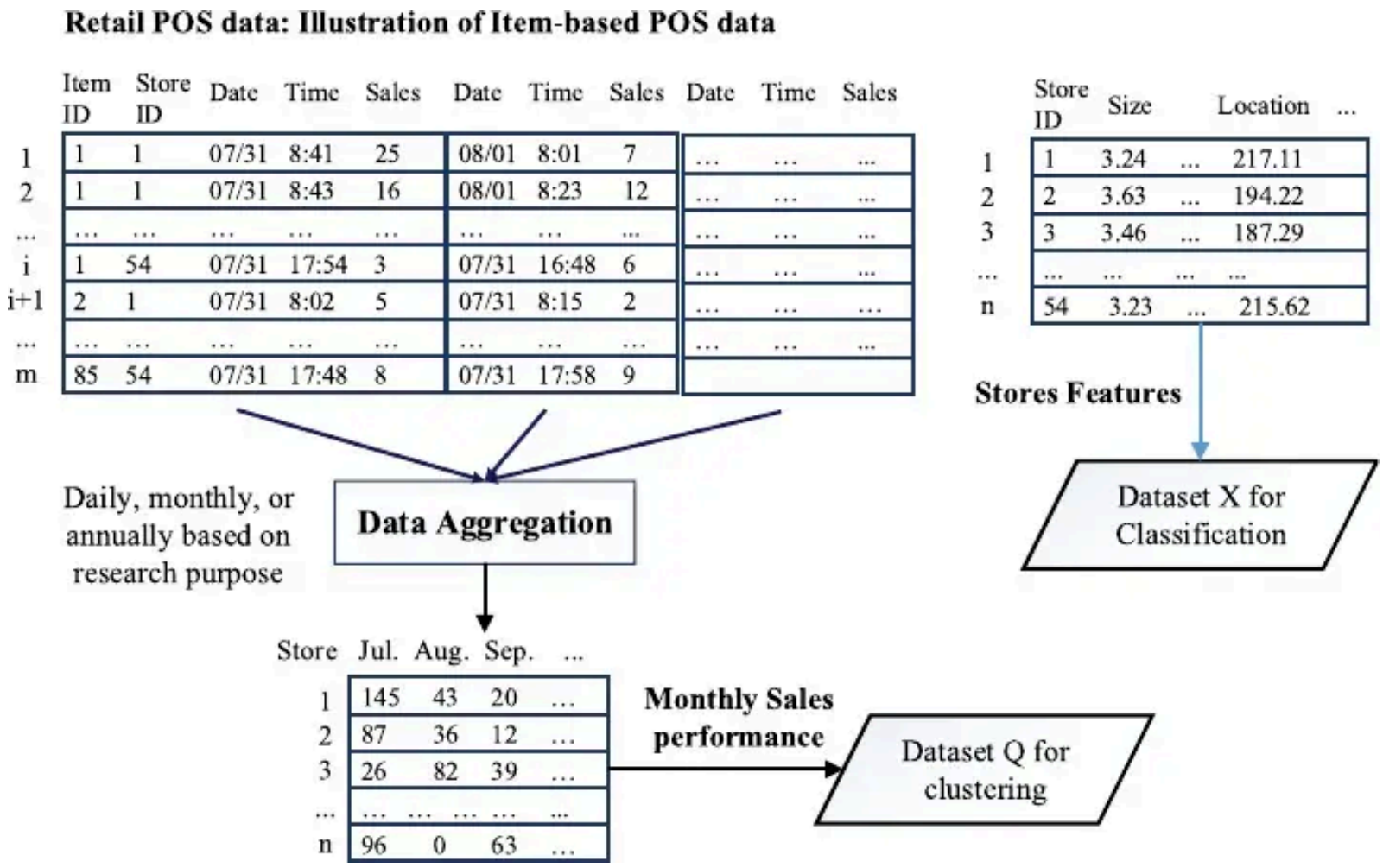


Fig. 2. Illustration of identifying data features for clustering and classi¯cation of retail POS data.

time durations such as daily or weekly without losing the generality. For example, in our case study, the Q dataset was aggregated as the total sales transactions per month of 54 retail bakery stores which sold more than 500 products in each store. The second dataset X can include store characteristics such as store geometry size, location of the store, and surrounding facilities nearby which the manager might be interested in ¯nding the association against the sales performance of the store. These two datasets Q and X will be used for clustering and classi¯cation, respectively, for investigating store features that correlate to the sales performance under certain groups of stores in terms of their sales performance. The process to identify data features for each dataset is illustrated in Fig. 2.

3.2. *Stage 2: Sequential clustering and classi¯cation based on NSGAII*

NSGAII-SCC, a sequential clustering and classi¯cation framework, proposed in Ref. 11 can be applied after separating the POS data into two Q and X datasets. Figure 3 illustrates the NSGAII-SCC framework with the proposed stepwise regression and TOPSIS methods. First, the initial population, which consists of two types of chromosomes, one for clustering and another one for classi¯cation, is generated. Feature selection is processed based on the chromosome design. Data clustering is ¯rst implemented based on the selected features on the Q dataset. The class labels obtained from the clustering result are then used to train the classi¯er. A multi-objective function, which considers the mean square error (MSE) and accuracy as clustering's and classi¯cation's performance measures, respectively, is calculated

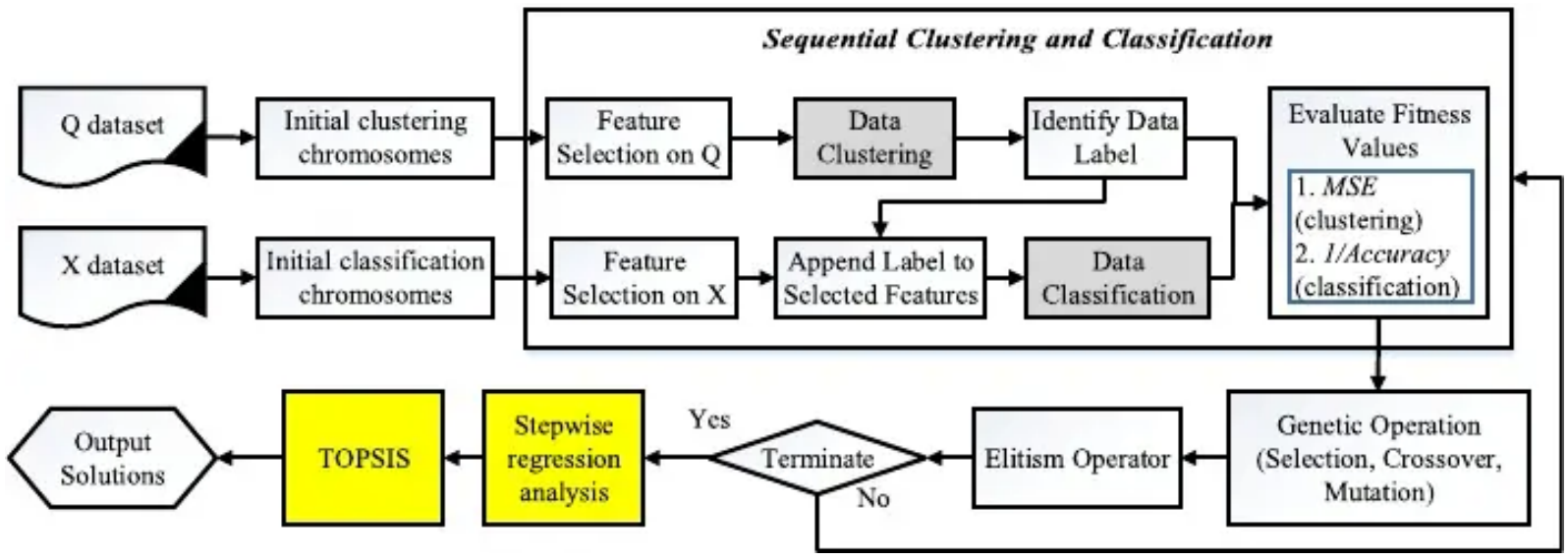


Fig. 3. Illustration of sequential clustering and classi¯cation data analysis framework with stepwise regression feature selection and determination of number of cluster by TOPSIS.

for each chromosome. Then, genetic operations such as selection, crossover, and mutation on feature chromosomes are then applied to search for the solutions for clustering and classi¯cation. Similar to the original NSGAII method which was proposed by Deb *et al.*,[38] non-dominated sorting and crowding distance are computed to select populations for the next generation. This process of the genetic algorithm (GA) will repeat until the stopping criterion is met. The output solutions on the Pareto front represent the solution candidates. In addition, the chromosomes of each generation are recorded for stepwise regression analysis to study factors on classi¯cation accuracy. The TOPSIS was then applied to select the solution for concluding the analysis. The details of NSGAII-SCC with stepwise regression feature selection and TOPSIS will be described in the following subsections.

3.2.1. *Chromosome representation*

Xue *et al.*[39] proposed a multi-objective-based NSGA for feature selection. In their method, a chromosome is encoded as a vector of real numbers in the range of [0,1] where the length of the vector is the number of features in the dataset. To identify which features are selected, the chromosome is transformed to a binary vector in which gene 1 denotes the selected feature while gene 0 represents an unselected feature. This study employs a similar approach in chromosome design. A chromosome is coded as a string of binary numbers in which each bit represents one data feature. Therefore, the length of a chromosome depends on the number of features in a dataset. For instance, in this study, clustering chromosomes are expressed as 12 bit-strings to correspond to 12 features in the sales dataset Q. One clustering chromosome sample is $f010010110011g$. In this case, the genes which contain the value $f1g$ appear on the 2nd, 5th, 7th, 8th, 11th, and 12th positions of the chromosome mean that the corresponding features on these positions in the sales dataset Q will be selected to perform clustering. This concept of chromosomes representation is applied similarly on the feature dataset X for classi¯cation.

3.2.2. *Sequential clustering and classi¯cation*

Feature selection

The feature selection mechanism is conducted by designing a chromosome in GA to indicate attributes that will be used for clustering and classi¯cation. The selected attributes from the sales dataset Q indicated by the chromosome are trained using a clustering method to reveal hidden data structures. The clustering result for each store is labeled in the store features dataset X. The selected attributes in dataset X indicated by the chromosome were used as inputs to classify the clustering labels for the classi¯cation process.

Sequential clustering and classi¯cation

After selecting data features from the chromosome set, clustering and classi¯cation algorithms are performed sequentially as shown in Fig. 3. The selection of clustering and classi¯cation methods in this combination model is an important research question. This study conducted a prior experiment on the given POS dataset based on the two clustering methods (hierarchical and k-means) and four classi¯cation methods, i.e., DT, ANN, KNN, and SVM, mentioned in the literature review in Sec. 2.2. The prior experiment result will be shown in Sec. 4.2.

Fitness function

The NSGAII-SCC framework assesses the performance quality of both clustering and classi¯cation tasks. For clustering, MSE is used to evaluate the compactness of clusters. Similarly, the classi¯cation performance is measured by the prediction accuracy (acc). On the one hand, the clustering task aims to minimize MSE which means that the clusters are more compact. On the other hand, classi¯cation seeks to maximize acc to increase prediction accuracy. To be consistent with the clustering performance index, the reciprocal of acc, which is denoted as $1/acc$, is used to measure the performance of classi¯cation. Therefore, the ¯tness function now considers minimizing both MSE and $1/acc$.

3.2.3. *Genetic operations*

To produce the next generation in genetic algorithm, genetic operations are conducted in the next steps. In common, genetic operations consist of selection, crossover, and mutation. This study follows the design of original NSGAII [38] for these tasks. For instance, the selection operation uses the tournament method to select chromosomes from the population. Crossover is conducted based on the conventional one-point crossover. Mutation operation changes one random binary gene from $j1j$ to $j0j$ or vice versa.

3.2.4. *Non-dominated sorting*

Non-dominated sorting procedure is a crucial task to evaluate the solutions of clustering and classi¯cation. The elitism non-dominated sorting combines populations so that the solutions are not trapped in a local optimum. The o®spring chromosomes, which are reproduced after implementing genetic operations, are applied

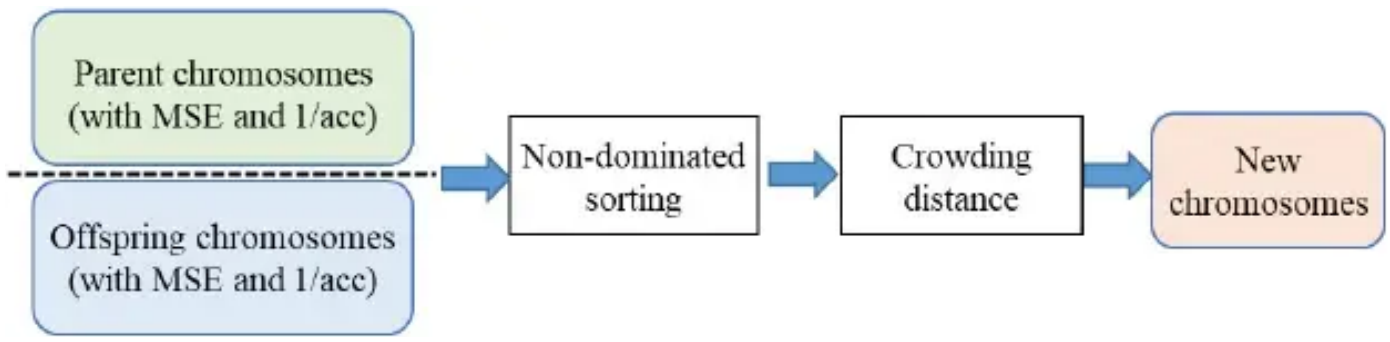


Fig. 4. Non-dominated sorting procedure of the NSGAII-SCC framework.

to the sequential clustering and classi¯cation algorithm to obtain the ¯tness values. Then, the procedure follows the original algorithm of Deb *et al.*[38] Parent chromosomes are combined with o®spring chromosomes to conduct non-dominated sorting and calculate crowding distance to select the next generation. Figure 4 illustrates the non-dominated sorting procedure of the NSGAII-SCC framework.[11]

The process of GA will repeat until the stopping criterion is met. The output solutions on the Pareto front represent the solution candidates.

3.2.5. *Feature selection by stepwise regression analysis*

After the optimization process is ¯nished, the algorithm can obtain the optimal results of the clustering and classi¯cation based on their ¯tness values. However, multiple solutions generated by NSGAII-SCC technique should be investigated to ¯nd the correlated factors from the X dataset that a®ect the revealed patterns. Thus, this step focuses on exploring the in°uence factor on classi¯cation accuracy based on the feature selection mechanism.

First, the chromosomes for classi¯cation of each generation are recorded during the iterative process of the sequential clustering and classi¯cation. Then, stepwise regression[15] is utilized to analyze the in°uent features. The stepwise regression method can detect each variable to eliminate the non-signi¯cant features and keep the appropriate ones for the ¯t model. Suppose that the population size and number of generations are 50 and 100, respectively. A total of 5000 chromosomes for classi¯cation are recorded. To build the stepwise regression models, the *acc* is de¯ned as a dependent variable while the features on the X dataset are de¯ned as independent variables. This technique takes a step-by-step approach to selecting variables. The process of adding or eliminating independent variables is implemented by using the statistical signi¯cance of the variables. A ¯nal model is adopted when no more independent variables can be added or deleted from the stepwise model. Figure 5 illustrates the framework of the stepwise regression mechanism on the chromosomes for classi¯cation. As can be seen, the numbers of 1/acc are dependent variables of the stepwise regression model while the features of X dataset are considered as the independent variables. After performing stepwise regression, the features selected will be kept for reasoning the relationship between X and Q datasets.

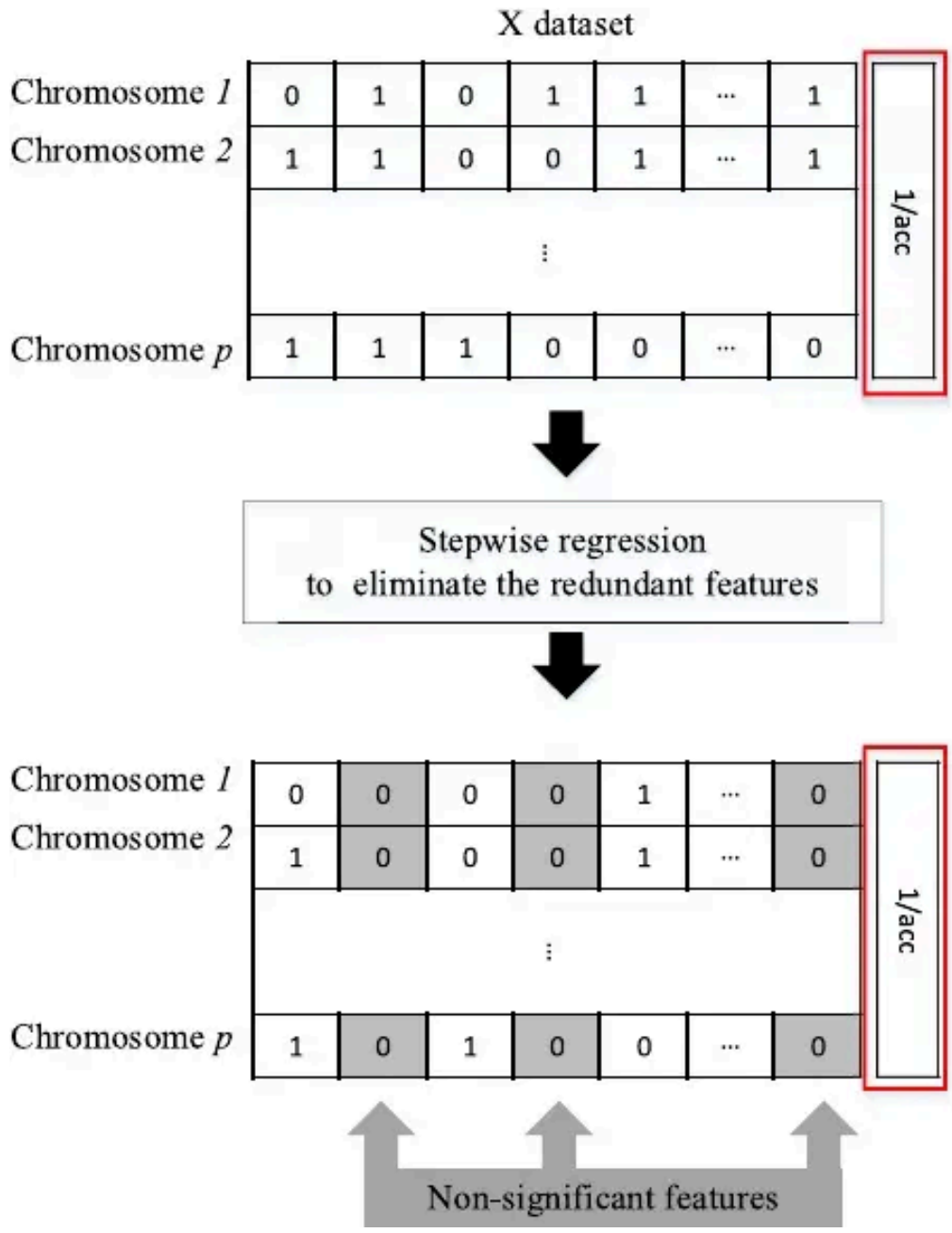


Fig. 5. The framework of stepwise regression with feature elimination.

3.2.6. *Determination of number of cluster by TOPSIS*

To determine the number of cluster k based on the Pareto front set, this study integrates the TOPSIS method to select the solution which has the largest relative closeness considering the weights of di®erent criteria. Essentially, TOPSIS is a multi-criteria decision analysis method to select the alternative that has the shortest geometric distance from the positive ideal solution and the longest geometric distance from the negative ideal solution. [40] A set of alternatives is compared by (1) identifying weights for each criterion, (2) normalizing scores for each criterion, and (3) calculating the geometric distance between each alternative and the ideal solution, which has the best score in each criterion.

In addition to the two objective functions MSE and 1/ acc, two more criteria are represented by the number of selected features in the non-dominated solutions. For each solution, the frequency proportion of the clustering chromosomes (FR-clus) is calculated by dividing the number of selected features by the total features in the clustering chromosome. Similarly, the frequency proportion of the classi¯cation chromosomes (FR-class) is the number of selected features divided by the total features in the classi¯cation chromosomes. The objective of considering FR-clus and

FR-class is to ¯nd out the solutions which have a relatively smaller number of features used based on the principle of Occam's razor in data science (simple is better). [41] After assigning weights for MSE, 1/ *acc*, FR-clus, and FR-class, TOPSIS method is used to determine the relative closeness of each solution. Then, the number of cluster associated with the solution which has the largest relative closeness will be determined as the cluster of X dataset. This determination is crucial for further analyzing the relationship between store performance and store features on POS data.

3.3. *Stage 3: Result analysis*

After determining the number of cluster, the further analysis can be conducted to investigate the optimal solution for POS analysis. Essentially, the constructed clusters can show the sales pattern in each group of stores. A similar seasonal e®ect on revenue can be easily revealed by the clustering results. The classi¯cation result will present store features that are correlated to the sales pattern for each group of stores. This analysis will be bene¯cial to improve sales performance and for decision-making purposes. In the next section, a real-world case study will be used to present the clustering and classi¯cation analysis combined.

4. Case Study

4.1. *Data description and parameter setting*

In this case study, POS data are collected from a bakery franchisee in Hangzhou, China. The data are divided into two sets, one for clustering and the other one for classi¯cation. The ¯rst set is the sales dataset Q that contains aggregated monthly revenue by store. There are 54 bakery stores and each store sells more than 500 products in this bakery franchisee. The second portion is the store features dataset X that comprises characteristics information related to each store. The factors included in the store features dataset are as follows:

- Store size: Store size by square meter.
- Floor: The Number of °oors in the stores (one, two, or three).
- Bus: The Number of bus stations nearby the bakery stores.
- Facility: Public facilities surrounding the stores such as MRT station, school, government o±ce, department store, convenience store, residential area, and tourist attraction spot. This is an attributing factor: 1 means a public facility exists and 0 means there is no public facility.
- Location: District location of the stores. The districts are symbolized by the numbers 1 to 10 as shown in Table 1.
- Population: Population in the districts where the stores are located at.
- Population density: Measurement of population per unit area.

User-de¯ned parameters for the proposed NSGAII-SCC are set up as follows: number of generations ¼ 100, population size ¼ 100, crossover rate ¼ 0.8, mutation

Table 1. Districts names and districts symbols of POS data.

Symbol	District name	Symbol	District name
1	Shang Cheng	6	Bin Jiang
2	Xia Cheng	7	Xiao Shan
3	Jiang Gan	8	Yu Hang
4	Gong Shu	9	Fu Yang
5	Xi Hu	10	Jia Xing

rate ¼ 0 1, weighting component ¼ 1 2 and chromosome selection ¼ 0 1. The number of generations is used as the stopping criteria for GA. The number of clusters (k) ranges from 2 to 10. To evaluate the performance of the NSGAII-SCC framework on the POS data, the results will be compared with the conventional sequential clustering and classi¯cation (SCC) without NSGA. As mentioned in the previous section, two validation indices (MSE and 1/ acc) are used to measure the results. The clustering result is validated by the MSE index and the classi¯cation performance is evaluated by 1/ acc.

4.2. *Prior experiment*

Several data mining techniques can be used to perform clustering and classi¯cation. This prior experiment aims to investigate the best combination of clustering and classi¯cation methods for this POS data analysis. For clustering, k-means and hierarchical agglomerative clustering are selected. Regarding classi¯cation methods, DT, ANN, KNN, and SVM are chosen. The prior experiment is conducted to the given POS data based on the conventional SCC method. Similar to the NSGAII-SCC framework, the conventional SCC ¯rst performs clustering on the sales data Q and gets data labels from the clustering result to conduct classi¯cation on store features dataset X. The experiment is repeated 20 times and the average MSE and 1/ acc results are recorded. The clustering results of k-means and hierarchical

Table 2. Results of clustering and classi¯cation based on SCC.

k	MSE (*10^8Þ		1/ acc			
	Hierarchical	k-means	ANN	KNN	SVM	DT
2	18.33	14.01	1.30	1.22	1.21	1.11
3	9.46	8.20	1.32	1.35	1.40	1.13
4	8.64	5.40	1.29	1.44	1.66	1.18
5	8.26	4.02	1.40	1.53	1.96	1.19
6	7.90	3.48	2.32	1.65	1.68	1.25
7	3.53	2.74	1.97	1.77	2.16	1.49
8	3.02	2.29	2.27	2.03	1.92	1.52
9	2.80	2.01	2.55	1.98	1.80	1.60
10	1.70	1.67	2.14	2.35	1.87	1.66

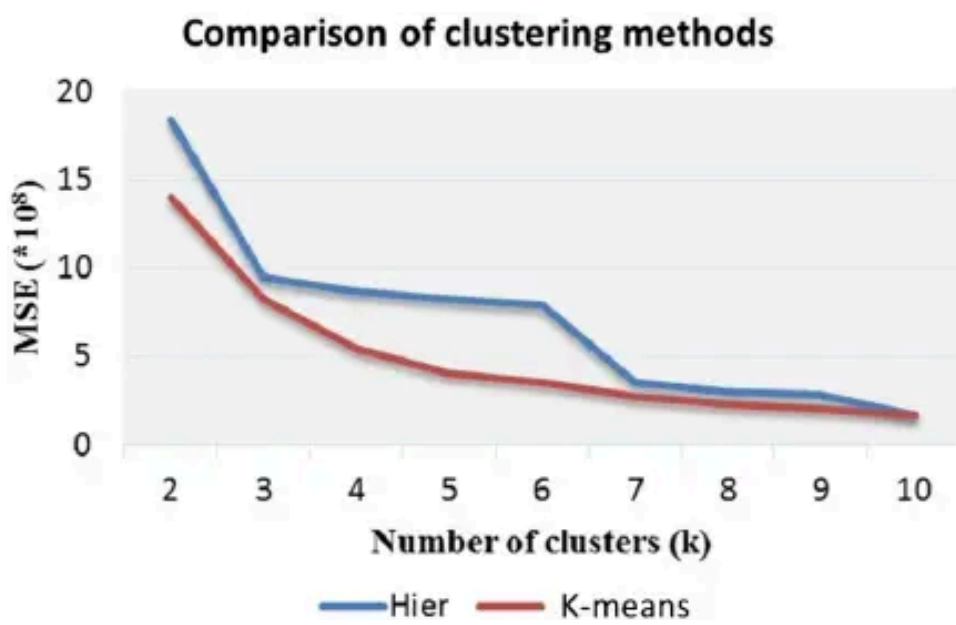


Fig. 6. Comparison of clustering methods based on bakery POS data.

agglomerative clustering are compared to select the best one. Similarly, ANN, KNN, SVM, and DT results are compared to choose the method that can give the best results on the given POS dataset. The results are shown in Table 2 and plotted in Figs. 6 and 7.

Figure 6 shows that *k*-means performs better than hierarchical clustering in all values of *k*. For two to six clusters, there is a signi¯cant di®erence in MSE between *k*-means and the hierarchical method. However, the di®erence decreases when the number of clusters exceeds six. Based on this result, *k*-means is selected to perform the clustering task on the given POS dataset.

Similarly, Fig. 7 illustrates the comparison of four classi¯cation methods using the POS data. DT performs better than other methods in terms of 1/*acc* (smaller is better). In most cases, the 1/*acc* index rises steadily when the number of clusters increases. This indicates that the classi¯cation technique becomes less accurate when there are more class labels (*k*).

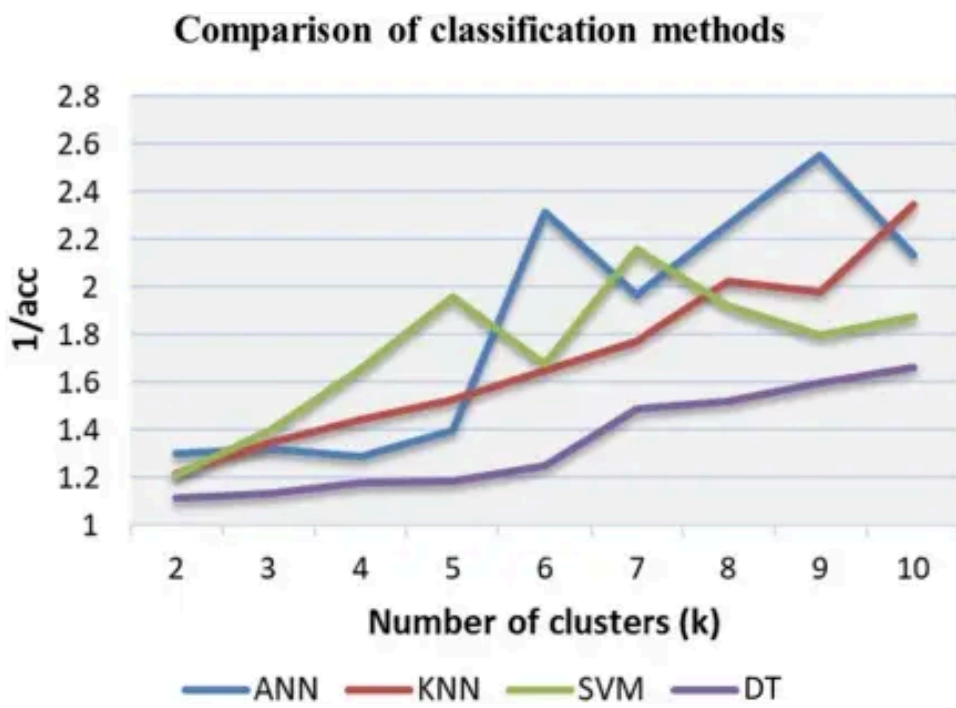
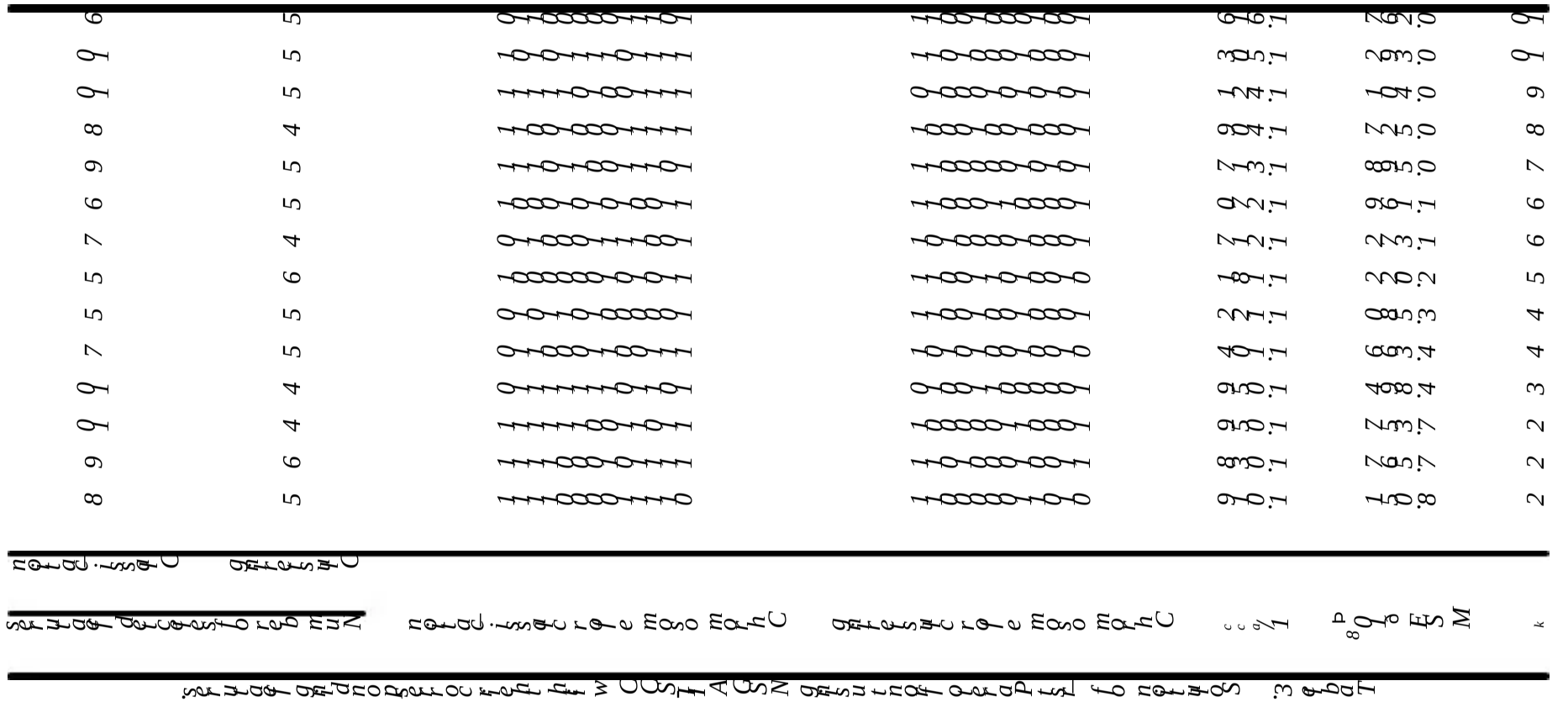


Fig. 7. Comparison of classi¯cation methods based on bakery POS data.


sequential decision tree
non-sequential decision tree
sequential greedy tree
non-sequential greedy tree
COSTA SONG
greedy sequential tree
non-sequential tree
sequential tree
Tabu Search
ACO
EM
k

Based on the results of this prior experiment, this study selects k-means and DT to perform clustering and classi¯cation tasks, respectively, for the POS data using the NSGAII-SCC framework. The results of NSGAII-SCC are compared with SCC to evaluate the performance of the proposed framework.

4.3. *Comparison of NSGAII-SCC and SCC results*

The proposed NSGAII-SCC framework provides a set of non-dominated solutions, also known as the Pareto front. In this experiment, the number of clusters k is assigned from 2 to 10. For each value of k, the clustering task is performed to obtain the corresponding label L. L is then used to train the classi¯er in the classi¯cation task. As the result, the NSGAII-SCC provides the Pareto frontier with di®erent values of k. Please note that, for each value of k, there are several non-dominated solutions, as shown in Table 3. The average result of NSGAII-SCC for each value of k is taken to make a comparison with the conventional SCC.

In Table 3, the Pareto front solutions with their corresponding chromosomes which are used to identify the selected features are exhibited. MSE and $1/acc$ are the validity indices of clustering and classi¯cation, respectively. Smaller values of MSE and $1/acc$ indicate the better result. Binary chromosome represents the data features \1" which means the feature is included in the training data while \0" signi¯es vice versa. As shown in Table 3, the binary indicates in chromosome specify the used features for conducting clustering and classi¯cation. Obviously, during the GA process, the multiple combinations of features selections will be evaluated. Only the mutual (also called Pareto) solutions will be presented as the candidates.

Table 4 shows the comparison of NSGAII-SCC and SCC frameworks in terms of MSE and $1/acc$. The clustering result of NSGAII-SCC is signi¯cantly better than SCC due to the smaller values of MSE. Also, NSGAII-SCC has a smaller $1/acc$ than SCC which means NSGAII-SCC has better classi¯cation accuracy. Based on the paired t-test with a 95% con¯dence interval on comparing NSGAII-SCC with SCC,

Table 4. Results of NSGAII-SCC and SCC frameworks based on bakery POS data.

k	NSGAII-SCC		SCC	
	MSE ($*10^8$)	$1/acc$	MSE ($*10^8$)	$1/acc$
2	7.66	1.04	14.01	1.11
3	4.89	1.06	8.20	1.13
4	3.97	1.11	5.40	1.18
5	2.02	1.18	4.02	1.19
6	1.27	1.24	3.48	1.25
7	0.60	1.32	2.74	1.49
8	0.53	1.41	2.29	1.52
9	0.40	1.42	2.01	1.60
10	0.33	1.56	1.67	1.66
mean	2.41	1.26	4.87	1.35

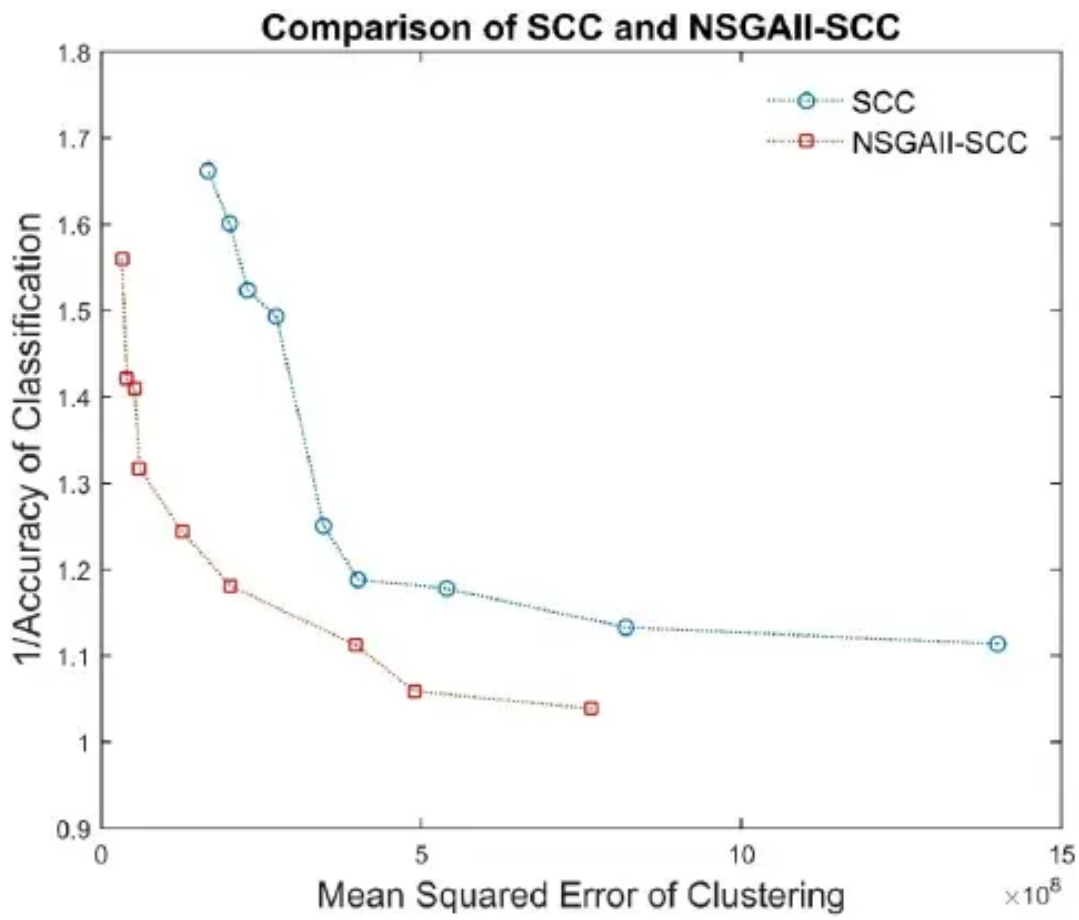


Fig. 8. Solutions of ¯rst Pareto front using NSGAII-SCC and SCC methods based on bakery POS data.

the statistical result con¯rmed the superior of NSGAII-SCC against SCC in terms of smaller MSE and 1/ *acc*.

Figure 8 further illustrates the solutions of NSGAII-SCC and SCC methods based on the bakery POS data. In Fig. 8, the red line with the square indicator shows the solutions of the ¯rst Pareto front using NSGAII-SCC, while the blue line with the cycle indicator shows the solutions of the ¯rst Pareto front using SCC. The red line with the square indicator is closer to the original coordinate (left-bottom corner) which means NSGAII-SCC can obtain the smaller MSE and 1/ *acc* simultaneously. Once again, this result shows that the proposed framework NSGAII-SCC outperforms SCC in terms of better clustering and classi¯cation performances. It also means NSGAII-SCC can search the better solutions of clustering and classi¯cation when conducting POS analysis for revenue performance investigation.

4.4. *Solution selection for the number of clusters k*

As mentioned in the methodology section, TOPSIS method was applied here to determine the number of *k* based on the solutions generated sequential clustering and classi¯cation. Table 5 lists the comparison of the number of clusters by the TOPSIS method calculation process. The ¯rst set contains bene¯t attributes for maximization (more is better). Another set comprises negative attributes for minimization (less is better). In this study, *acc* is selected as the bene¯t attribute while MSE, FR-clus, and FR-class are the negative attributes. All attributes' scores are normalized and presented in Table 5. TOPSIS method calculates the relative closeness to the ideal solution and then ranks the solutions. As can be seen in Table 5, the ¯rst ranked solution is selected, which is the solution with *k* ¼ 5. It also means that while cluster

Table 5. Comparison of number of cluster (*k*) by TOPSIS method.

Weight	0.3	0.2	0.2	0.3	Relative closeness	Ranking
k	MSE	FR-clus	FR-class	acc		
2	1.000	0.417	0.615	1.000	0.438	11
2	0.938	0.500	0.692	0.949	0.436	12
2	0.911	0.333	0.769	0.898	0.435	13
3	0.594	0.333	0.769	0.898	0.564	10
4	0.527	0.417	0.538	0.792	0.585	8
4	0.426	0.417	0.385	0.752	0.643	5
5	0.225	0.417	0.462	0.629	0.720	1
6	0.142	0.333	0.538	0.559	0.713	2
6	0.116	0.417	0.462	0.465	0.688	3
7	0.042	0.417	0.692	0.387	0.662	4
8	0.033	0.333	0.615	0.250	0.625	6
9	0.017	0.417	0.769	0.234	0.611	7
10	0.016	0.417	0.769	0.129	0.581	9
10	0.000	0.417	0.462	0.000	0.564	10

revenue performance of stores, ¯ve groups of stores can be established for better analysis due to the relevant clustering and classi¯cation results combined in Pareto front solutions.

4.5. *POS data analysis*

According to the result from the TOPSIS method, the solution with ¯ve clusters ($k = 5$) is selected for POS data analysis. In this case, 54 bakery franchise stores are grouped into ¯ve clusters by the monthly revenue. Figure 9 plots the sales revenue of

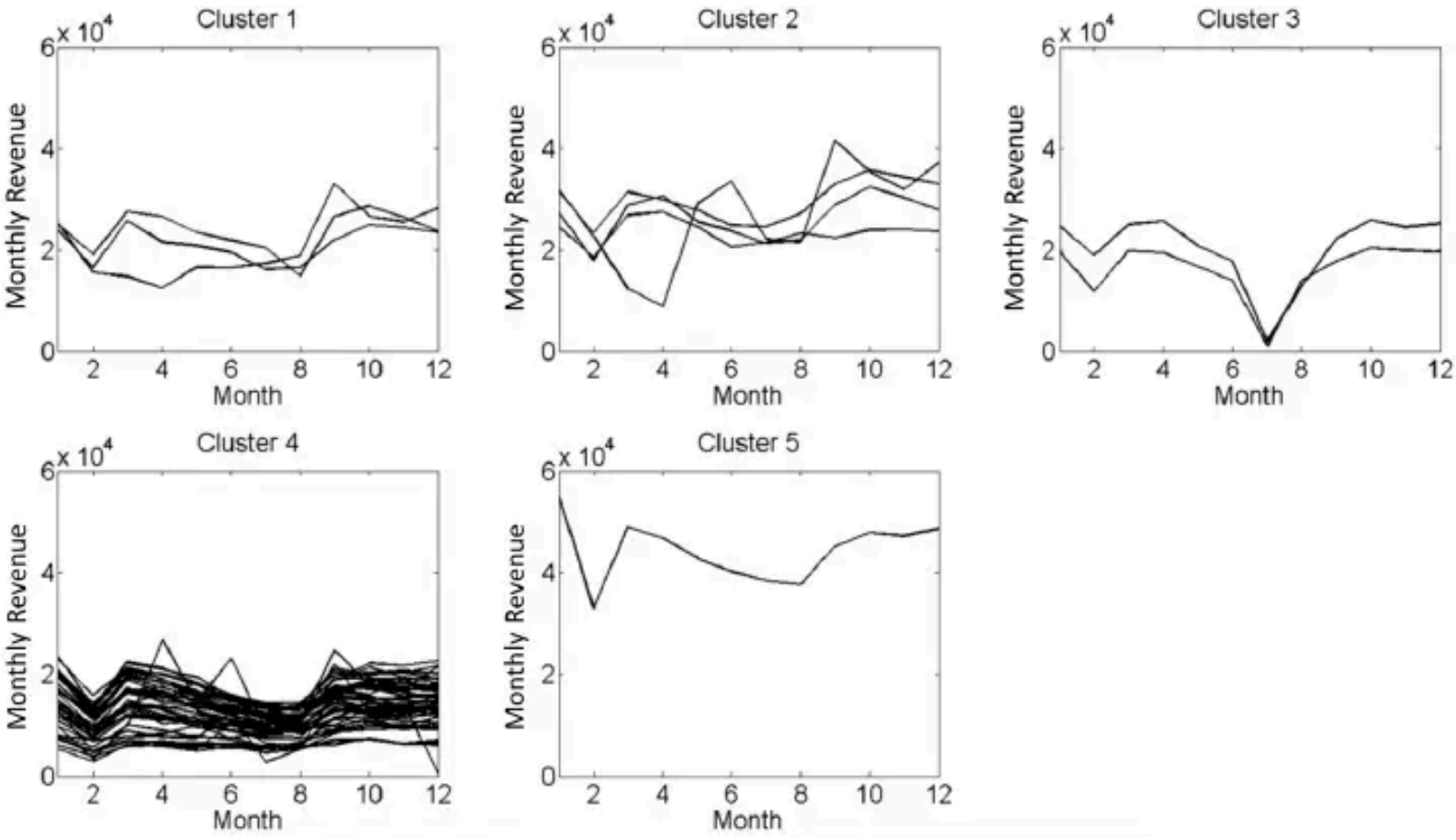
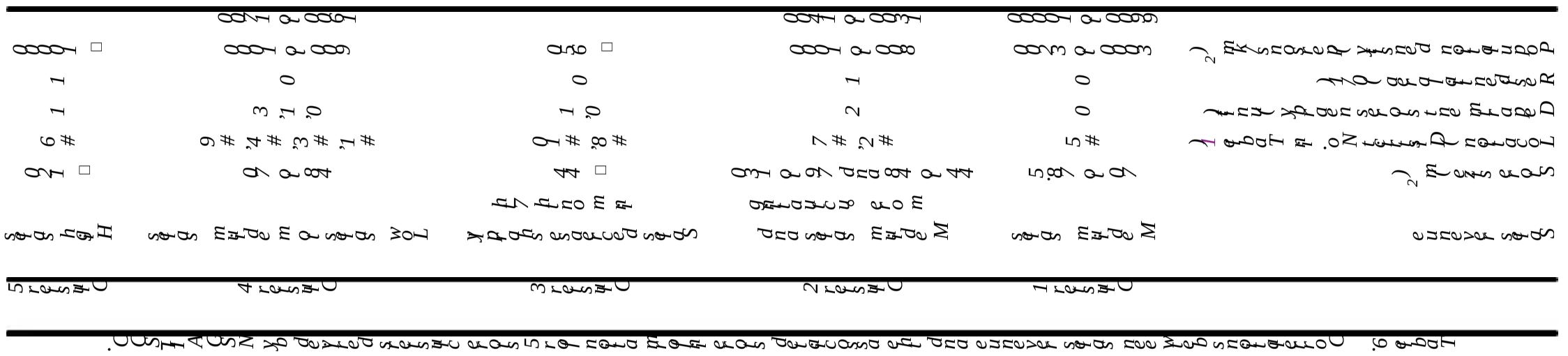


Fig. 9. Monthly revenue of each group of stores.

each cluster to show the sales patterns among stores. Similar sales patterns in one cluster can be recognized. For example, most of the stores are grouped into Cluster 4 which had diminished revenue in the 2nd, 6th, and 8th months. Meanwhile, Cluster 5 has only one store that has outstanding revenue each month. Cluster 3 contains stores with a signi¯cant decrease of revenue at the 7th month which shows the unique patterns than other groups. Cluster 1 and Cluster 2 seem not much distinct from each other but Cluster 2 has relatively larger variation. Based on this clustering analysis, the groups of stores with di®erent sale patterns can be revealed for further investigation.

The chromosomes which represent the selected features help us to identify important features for both clustering and classi¯cation. Following the selected solution at $k = 5$, the clustering chromosome is a string of $f010010110011g$. This indicates that six features, among a total of 12 features in the revenue dataset, are chosen as the representative features to perform clustering. These six selected features are the revenue of the 2nd, 5th, 7th, 8th, 11th, and 12th months. These six months are signi¯cant in di®erentiating retail stores shown in Fig. 9 because they represent the critical sales performance when conducting a comparison of all stores. For instance, the 7th feature contributes to the signi¯cant pattern characteristic in Cluster 3. The 2nd feature is signi¯cant in most of the clusters. Therefore, using the selected features can predict clustering results that lead closer to the result while using all data. It also means that the proposed framework provides not only the mechanism of determining cluster number considered clustering and classi¯cation result but also selecting the features which are the most in°uential for store clustering.

Once the clustering method obtains the store labels, the store features are classi¯ed using the clustering labels as classi¯cation targets. Similar to the process of exploring selected features for data clustering, the chromosome for classi¯cation corresponding to the optimal value of k determined by TOPSIS is analyzed. Besides, the stepwise regression method is applied on the classi¯cation chromosomes to eliminate the redundant and non-signi¯cant features to the *acc*. The signi¯cance level is set as 0.1. Finally, the ¯ve most correlated factors, i.e., store size, department stores, residential area, location, and population density, can be used to classify the store cluster. The correlations between sales revenue and the associated store information for ¯ve store clusters are described in Table 6. As can been seen, store size, department stores, residential area, location, and population density of each store cluster have di®erent magnitude levels. For example, Cluster 5 was identi¯ed as the high sales revenue store which happens to have a larger store size in a high population density region. However, the stores with bigger °oor plans do not guarantee tremulous sales. There is no negligible °uctuation in sales in Cluster 4; even the stores in that cluster have middle store size. Based on the analysis result, the department stores nearby turn out to be a signi¯cant factor to identify the sale performance in Cluster 4. Moreover, Cluster 3 can be recognized as a special cluster which is located in the lowest population density region #8 and #10 and relatively low sales in month 7th. This analysis points out the stores which were located near


Low Head Pressure
Head Pressure Control
Pressure Control
High Head Pressure
Head Pressure Control
Pressure Control
Head Pressure Control
Pressure Control
Head Pressure Control
Pressure Control

elementary schools in suburban areas. It can be easily related to the low sales during summer because students do not need to go to school during the break. Please note that although the analysis seems intuitive, the features (factors) mentioned above were found out by the proposed algorithm without any human intervention. The analysis afterward actually shows that NSGAII-SCC can cluster the stores by sales revenue and classify relevant features which correlate signi¯cantly to the clustering result (monthly revenue). Last but not the least, the result shown in Table 6 demonstrates the useful information that came out by the framework for the retailing store manager to appraisal the performances of the store clusters based on the provided POS data.

4.6. *Discussion*

In this research, the NSGAII-SCC was compared with the conventional SCC method which has been widely used for data mining. Without considering the complexity of the combination of clustering and classi¯cation, SCC has straightforward analytics which considers clustering and classi¯cation as the separated tasks. For the analysis scenarios of retail revenue performance, the determination of the number of store clusters and the feature selection for both clustering and classi¯cation are confounded together. It means that for any number of clusters determined ¯rst, the signi¯cant features for clustering and classi¯cation might be di®erent to guarantee the analysis's e®ectiveness. Also, even we can select features ¯rst for clustering and classi¯cation, the performance under the di®erent number of clusters might be diverse. By using the multi-objective framework with GA searching operations, the solutions on the Pareto front provide a relatively smaller number of solution candidates for further investigation. The solutions also have non-dominated characteristics to ensure the e®ectiveness of clustering and classi¯cation combined.

Without a doubt, NSGAII-SCC needs more computational e®ort because of the iterative processes under the GA framework. Generally speaking, NSGAII-SCC needs more than 20 ¯100 times of computational time than SCC depending on how the stop criteria of GA is set. However, this computation framework can be executed automatically under the database platform as the stored procedure, or the analytics platform with data mining subroutines. The analytics framework of NSGAII-SCC is paid o® without exhaustively searching all combinations with a number of $\sum_{p=1}^{k}\binom{n}{p}\binom{m}{p}$, where k is the largest number of clusters; n is the number of features for clustering; m is the number of features for classi¯cation. For example, in the cast of POS study shown in Table 3, the NSGAII-SCC can generate Pareto front without going through $\sum_{p=1}^{10}\binom{12}{p}\binom{13}{p} = 301{,}879{,}305$ combinations of clustering and classi¯cation with k=2 ¯10. This meta-heuristic method proposed by NSGAII-SCC can still be considered computational e±ciency.

The setting of weights in the TOPSIS method also provides the capability of managerial adjustment. If the manager of a retail store has more concerned about the

accuracy of classi¯cation results, the weight of *acc* can be adjusted higher. Then, the ranking of choosing the number of store clusters will be tended to more favorable to the one which has better classi¯cation accuracy. Similarly, if the manager would like to ensure the clustering has better compactness, the weight of MSE can be set higher comparing with other decision variables.

In addition, the stepwise regression method is employed to eliminate the redundant and non-signi¯cant features that are correlated to the store patterns. The removed features may depend on the signi¯cance level. If the *p*-value of the *t*-test in the stepwise model is greater than the signi¯cance level for removal, the feature will be eliminated from the stepwise model. Thus, setting the higher value for signi¯cance level may lead to reduce the probability of removing a feature from the stepwise model. In contrast, the signi¯cant level can be set as a small value to increase the probability of features' removal.

5. Conclusion and Future Research Direction

Analyzing retail POS data to understand the store's performance is critical for retail management. By investigating the pattern of sales, the management can perform better decision-making on handling the inventory or adjust the list of items. In this research, the NSGAII-SCC framework was applied to discover the hidden sales pattern of stores in terms of sales revenue and further reveal the correlation between the sales revenue and store characteristics by classi¯cation method. NSGAII-SCC framework constructs the multi-objective optimization to maintain clustering and classi¯cation results in terms of compactness of clusters and classi¯cation accuracy, respectively. Through the recursive procedures of GA operation, the chromosome settings can represent the feature selection evolved to search for the optimal clustering and classi¯cation result and select the correlated data features. The solutions of the Pareto front can be plotted to show the population of solutions found. Three analysis operations including (1) the determination of the number of store clusters, (2) the feature selection in dataset *Q* for clustering store revenue performance, and (3) the feature selection in dataset *X* for classifying the store cluster can be performed at once by the proposed NSGAII-SCC framework.

In order to conclude the multiple solutions from Pareto sets generated by NSGAII-SCC, this work integrates the stepwise regression and TOPSIS method to determine the signi¯cant features from store characteristics and the number of cluster in store revenue, respectively. The integration of the stepwise regression and TOPSIS can enhance the usability of NSGAII-SCC to provide the post-data analysis. This integration also extends the NSGAII-SCC framework to resolve the solution and feature selection problem when the extreme large number of solutions might exist when applying it on the real-world data.

To demonstrate the application of the NSGAII-SCC framework, the POS data of a bakery retail chain are used as a case study to evaluate the proposed framework. First, the revenue dataset derived from POS data was denoted as the sales dataset *Q*

that measures stores' performance. Then, the relevant store information for each store is de¯ned as the store features' dataset X for the classi¯cation process. The proposed data analysis framework is substantial to investigate the store sales performance and reveal the correlated factors. This analysis can be performed by NSGAII-SCC automatically and the result has signi¯cant usage for improving sales and developing business strategies. Additionally, the proposed framework can explicit the best number of clusters account for the better clustering and classi¯cation performance using POSSIS. Finding out factors that correlate to the sales performance based on the stepwise method can further provide managerial meaning when studying the POS data.

For future research, there are multiple directions to improve the analysis and the algorithm framework. First, this result can be compared with other multivariate analysis methods. Second, the algorithm framework can be integrated with the standard database or ERP system to verify its performance. Third, a data preprocessing technique can be developed to screen out unimportant features either in the sales dataset or the store features' dataset to improve the computation e±ciency.

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